

WARSAW, IN, USA

WMO: 720266

Lat: 41.275N

Lon: 85.840W

Elev: 259

StdP: 98.25

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 54809

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-15.0	-22.4	0.5	-17.1	-19.7	0.7	-14.3	11.0	-3.4	9.8	-3.8	2.8	250	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.3	23.1	31.1	22.6	29.2	21.7	25.2	30.1	24.1	28.6	23.2	27.6	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.7	19.1	28.0	22.6	17.8	26.8	22.1	17.4	26.2	77.8	29.9	74.0	29.0	70.4	27.6	37.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.2	7.3	DB	-22.1	34.1	3.7	2.1	-24.8	35.6	-27.0	36.8	-29.1	38.0	-31.8	39.5
			WB	-22.4	26.3	3.8	1.4	-25.2	27.4	-27.4	28.2	-29.5	29.0	-32.3	30.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-3.7	-2.5	3.4	9.8	16.1	21.1	22.7	21.6	18.4	11.6	4.8	-0.6		
	DBStd	10.61	6.47	6.24	6.18	5.24	5.01	3.45	3.21	2.93	4.19	5.09	5.16	5.53		
	HDD10.0	1616	424	352	222	67	8	0	0	0	1	41	171	329		
	HDD18.3	3378	682	584	464	260	104	13	4	7	49	218	407	586		
	CDD10.0	1719	1	2	18	61	196	332	392	358	253	90	14	2		
	CDD18.3	438	0	0	1	3	34	94	138	107	52	8	0	0		
	CDH23.3	3875	0	0	10	34	346	801	1273	885	456	70	1	0		
	CDH26.7	1167	0	0	1	3	96	244	428	236	145	14	0	0		
(14) Wind	WSAvg	2.8	3.6	3.5	3.4	3.7	2.8	2.3	1.9	1.7	1.9	2.7	3.0	3.3		
(15) Precipitation	PrecAvg	999	69	47	59	89	111	104	111	112	77	83	76	60		
	PrecMax	1333	146	118	117	177	179	191	190	213	158	206	135	149		
	PrecMin	808	22	15	18	10	40	50	54	58	25	24	21	17		
	PrecStd	122	36	27	23	38	46	42	42	43	42	50	26	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	16.3	24.8	27.1	32.0	32.8	34.8	32.5	34.1	29.8	21.4	15.9		
		MCWB	12.3	11.8	17.3	16.3	21.3	22.4	24.8	23.6	23.5	20.4	13.7	13.5		
	2%	DB	11.1	12.6	19.9	23.8	29.0	31.3	32.3	31.1	30.2	25.9	17.6	12.4		
		MCWB	9.7	10.5	14.6	15.2	19.7	22.0	23.9	23.2	22.4	18.3	12.8	10.8		
	5%	DB	7.7	8.8	16.3	22.1	27.4	29.6	30.9	29.0	27.8	22.5	15.2	9.0		
		MCWB	6.2	6.9	11.7	14.2	18.7	21.5	23.1	22.3	20.5	16.8	11.2	7.4		
	10%	DB	5.0	5.8	12.9	19.0	25.0	27.6	28.9	27.6	26.1	20.1	12.7	7.0		
		MCWB	3.3	3.5	9.6	12.8	17.9	20.4	22.1	21.5	19.6	15.6	9.1	5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	13.6	17.5	18.2	22.7	25.1	26.8	25.8	27.6	22.2	15.8	13.9		
		MCDB	13.8	15.1	24.5	23.5	28.8	30.5	30.5	29.6	29.0	27.6	18.6	15.7		
	2%	WB	9.7	10.6	15.2	16.6	21.3	23.8	25.3	24.6	23.3	19.9	14.0	10.9		
		MCDB	10.6	12.4	19.0	21.4	26.8	28.7	30.5	28.7	28.5	23.4	16.8	11.9		
	5%	WB	6.5	6.9	12.8	15.2	20.0	22.9	24.4	23.7	22.2	18.1	11.7	7.6		
		MCDB	7.5	8.8	15.6	19.7	25.2	27.5	29.2	27.6	26.6	20.9	14.9	8.5		
	10%	WB	3.3	3.5	10.1	13.7	19.0	21.9	23.3	22.7	21.0	16.2	9.7	5.3		
		MCDB	4.9	5.4	13.1	19.2	23.6	25.9	27.6	26.5	23.8	19.5	12.8	6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	7.9	9.4	11.3	11.3	10.8	10.9	10.9	11.5	10.3	8.4	6.4		
		MCDBR	9.2	11.4	13.4	14.2	13.7	12.6	11.8	12.3	13.9	13.1	11.7	9.4		
	5% WB	MCWBR	8.3	9.4	9.0	7.9	6.7	5.5	4.9	5.6	6.8	7.1	8.1	8.2		
		MCWBR	9.1	11.1	12.1	12.7	11.5	10.0	10.3	9.7	11.7	10.9	10.4	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.326	0.349	0.400	0.441	0.436	0.442	0.446	0.409	0.352	0.318	0.296		
		taud	2.401	2.315	2.350	2.219	2.151	2.212	2.200	2.180	2.260	2.421	2.465	2.444		
	Ebn at Noon		859	878	899	870	839	843	833	817	822	840	826	828		
		Edh at Noon	80	102	111	136	150	141	142	140	121	92	76	72		
(44) All-Sky Solar Radiation	RadAvg		1.54	2.39	3.53	4.64	5.32	6.02	6.08	5.33	4.38	2.90	1.83	1.29		
	RadStd		0.15	0.31	0.31	0.41	0.50	0.39	0.33	0.29	0.36	0.28	0.25	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page